

Problem 41:

a)



5 interior

b)



6 exterior

c)



Problem 42:

a)



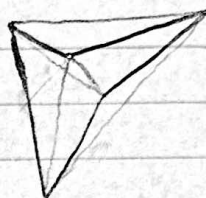
0 diagonals outside

b)

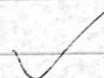


1 outside

c)



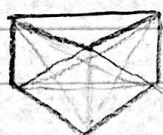
2 outside



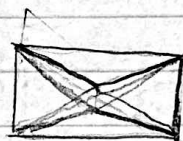
Problem 43:



0 diagonals outside



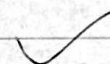
1 outside



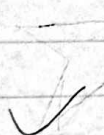
2 outside



6 outside



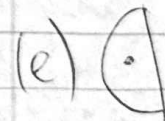
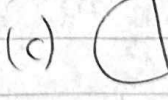
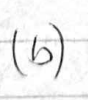
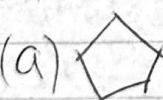
4 outside



Problem 44: (a) 11 (b) 9 (c) 7

On book

Problem 45:



a) on book

c) on book

e) on book

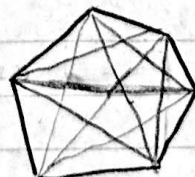
b) on book

d) on book

Problem 46:

a) 9 diagonals

b) 25



Problem 47:

lines

domains

$$\rightarrow \binom{n}{2} + 1$$



$$n=7 \checkmark$$

Problem 48: $(1,2), (1,3), (1,4), (1,5), (1,6), (1,6), (2,3), (2,4), (2,5),$

$(2,6), (2,7), (3,4), (3,5), (3,6), (3,7), (4,5), (4,6)$
 Problem 49: $(4,7), (5,6), (5,7), (6,7)$ 21 pairs $= \binom{7}{2}$ ✓

there are n neighbors, so its $\binom{n}{2} - n$

Problem 50:

$\binom{n}{2} - n$, it's the same as problem 49 but with a polygon ✓

Problem 51:

$\binom{n}{2}$, it's like problem 50 but you add the n sides ✓

Problem 52:

$$\begin{aligned} g &= P_3 \vee O \vee V_3 & a &= P_1 \vee P_3 \\ e &= V_1 \vee P_1 \vee O & b &= V_1 \vee V_3 \\ f &= V_2 \vee P_2 \vee O \\ t_1 &= P_2 \vee A \vee P_1 \\ t_3 &= P_2 \vee P_3 \vee C \\ t_2 &= V_1 \vee V_3 \vee A \\ t_4 &= V_2 \vee V_3 \vee C \end{aligned} \quad \checkmark$$

Problem 53:

Fig. 1.99 ✓